Scalability & Future Plan

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	1 Mark

This document outlines how the current solution can be scaled to handle larger user bases or extended features, and captures the team's roadmap for enhancing the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Public deployment runs BiLSTM-only with no real AI response.	Evaluators see a materially reduced experience compared to the local full mode.	High
2	CSV-based logging isn't durable on non-persistent hosting disks.	Interaction history resets whenever the Streamlit Cloud instance restarts.	Medium
3	No retrieval-based prediction path exists yet.	Can't reuse previously-seen similar examples for faster or cheaper repeat predictions.	Low
4	Single shared LLM API key (no bring-your-own key).	One user's usage counts against the same free-tier quota as everyone else.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Streamlit process, no load balancing.	Move to a container-based host with multiple replicas behind a load balancer if usage grows.
2	Data Storage	CSV file on a single machine.	Migrate interaction history to a lightweight managed database (e.g. SQLite → Postgres) once persistence across restarts matters.
3	Performance	Full-mode requests are I/O-bound on the LLM API call (see Performance Testing).	Consider response caching for repeated/similar prompts, and resolve the OpenRouter hang so full mode can run publicly.
4	Security	No authentication; open access.	Add basic auth or an API-key gate if the tool moves beyond a demo/assessment context.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Resolve the OpenRouter hang; restore full mode (BiLSTM + BERT + real AI) on the public deployment.	Short-term (2–4 weeks)	Evaluators and users get the complete experience, not the degraded one.
Phase 3	Build the CSV-based retrieval prediction pipeline properly, with a defined matching strategy.	Medium-term	Faster, cheaper responses for previously-seen problem patterns.
Phase 4	Refactor LLM clients away from module-level singletons; add bring-your-own API key.	Medium-term	Removes the shared free-tier bottleneck for multi-user scenarios.